

Xubo Luo

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Homepage: <https://luoxubo.github.io>

EDUCATION

Hong Kong University of Science and Technology

Hong Kong

PhD in Mechanical Engineering

Sep. 2026 - Present

- GPA: NA/4.00; work with Prof. *Yang Gao*
- Research interests: robot learning, reinforcement learning, navigation

University of Chinese Academy of Sciences

Beijing, China

M.S. in Computer Applied Technology

Sep. 2023 - Present

- GPA: 3.61/4.00; work with Prof. *Xue Wan* and Prof. *Wei Zhang*
- Relevant coursework: Deep Learning(91), Pattern Recognition and Machine Learning(92)
- Research interests: robotics vision, vision-language navigation, visual localization, and SLAM

Shanghai University of Finance and Economics

Shanghai, China

B.E. in Computer Science and Technology

Sep. 2019 - Jun. 2023

- GPA: 3.48/4.00 (Rank: 7/55); work with Dr. *Tsz Chiu Kwok*
- Relevant coursework: Linear Algebra(94), Discrete Mathematics(98), Data Structure(91), Computer Network(93)
- Research interests: image fusion, deep learning

SELECTED PUBLICATIONS

1. **X. Luo**, L. Wang, J. Zhang, D. Niu, "HBANet: A hybrid boundary-aware attention network for infrared and visible image fusion." *Computer Vision and Image Understanding (CVIU)*, 2024. [[PDF](#)] [[Code](#)]
2. **X. Luo**, Y. Tian, X. Wan, J. Xu, T. Ke, "Deep learning based cross-view image matching for UAV geo-localization." *International Conference on Service Robotics (ICoSR, oral)*, 2022. [[PDF](#)] [[Code](#)]
3. **X. Luo**, X. Wan, Y. Gao, Y. Tian, W. Zhang, L. Shu, "JointLoc: A Real-time Visual Localization Framework for Planetary UAVs Based on Joint Relative and Absolute Pose Estimation." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS, oral)*, 2024. [[PDF](#)] [[Code](#)]
4. **X. Luo**, Z. Zhao, S. Zhang, X. Wan, W. Zhang, L. Shu, "DVD-MapAL: Deep Visual Feature-Driven Map-Assisted Localization for Planetary Spacecraft Exploration." *IEEE Transactions on Field Robotics (T-FR)*, 2025. [[PDF](#)] [[Code](#)]
5. **X. Luo**, S. Zhang, X. Wan, W. Zhang, L. Shu, "Learning to Anchor Visual Odometry: KAN-Based Pose Regression for Planetary Landing." *IEEE Robotics and Automation Letters (RA-L, oral)*, 2025. [[PDF](#)] [[Code](#)]
6. **X. Luo**, A. Wu, H. Han, X. Wan, W. Zhang, L. Shu, "StepNav: Structured Trajectory Priors for Efficient and Multimodal Visual Navigation." *IEEE International Conference on Robotics & Automation (ICRA)*, 2026. [[PDF](#)] [[Code](#)]

RESEARCH EXPERIENCE

Shandong University Jul. 2020 - Aug. 2022

Research Assistant to Prof. Caiming Zhang and Prof. Xiuyang Zhao

- Developed algorithms for artifact removal in CT images, improving image quality.
- Implemented multi-modal image fusion techniques (infrared & visible, CT & MRI) to enhance diagnostic accuracy.
- Assisted in medical image segmentation tasks, contributing to the precision of image analysis.

Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences Sep. 2022 - Aug. 2023

Research Assistant to Prof. Xue Wan

- Conducted visual localization in earthquake-affected regions using remote sensing images.
- Assisted in the development of a multi-modal benchmark for lunar surface exploration.
- Developed a geography-based visual localization algorithm for drones, enhancing positioning accuracy.

AWARDS & HONORS

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|---|------|
| • Third Prize Scholarship, Shanghai University of Finance and Economics | 2020 |
| • ICPC Regional Competition Honorable Mention, ACM-ICPC | 2021 |
| • Honor Graduate, Shanghai University of Finance and Economics | 2023 |
| • Student Travel Grant, IEEE Robotics and Automation Society | 2024 |
| • First Place in CVPR AI4Space Workshop Challenge, IEEE | 2024 |
| • Merit Student, University of Chinese Academy of Sciences | 2025 |
| • First Place in IROS RoboSense Challenge, IEEE/RSJ | 2025 |
| • National Scholarship, Ministry of Education of China | 2025 |
| • Outstanding Graduate, University of Chinese Academy of Sciences | 2026 |

PROFESSIONAL SERVICES

Reviewer to:

- IEEE Robotics and Automation Letters (RAL 2024-2026)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE/SICE International Symposium on System Integrations (SII 2025, 2026)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025, 2026)
- IEEE International Conference on Robotics & Automation (ICRA 2026)

SKILLS

- Programming languages & Software: C/C++, MATLAB, Python, Linux, etc.
- Languages: Mandarin Chinese (Native), English(IELTS 7.0)